

VALENCE BOND THEORY

JEE Advanced Preparation - Chemistry

Sigma Bonds, Pi Bonds, Resonance & Limitations

Topic Overview

Valence Bond Theory (VBT) explains covalent bond formation in terms of overlap of atomic orbitals. It distinguishes sigma and pi bonds, explains multiple bonds, and lays the groundwork for understanding molecular geometry and reactivity.

1. POSTULATES OF VALENCE BOND THEORY

- **Postulate 1:** A covalent bond is formed by the **overlap of two half-filled atomic orbitals**, each from one combining atom.
- **Postulate 2:** The two electrons forming the bond must have **opposite spins** (one spin-up, one spin-down) — Pauli's principle.
- **Postulate 3:** The greater the overlap between two orbitals, the **stronger** the covalent bond formed.
- **Postulate 4:** Only valence shell orbitals participate in bond formation.
- **Postulate 5:** Each orbital can participate in only one bond (unless hybridised — then the hybrid orbital forms one bond).
- **Postulate 6:** The direction of the bond is along the region of maximum orbital overlap.

2. SIGMA (σ) BOND

A sigma bond is formed by **head-on (axial) overlap** of atomic orbitals along the internuclear axis. It is **cylindrically symmetrical** about the bond axis — i.e., rotation about the bond axis does not change the electron density distribution.

Types of Sigma Bond Overlap:

- s-s overlap: Both atoms contribute an s orbital. e.g., H-H in H_2
- s-p overlap: One atom contributes s, other contributes p orbital along axis. e.g., H-F in HF, H-Cl in HCl
- p-p overlap (head-on): Both atoms contribute p orbitals along the same axis. e.g., F-F in F_2

Key Property: Free rotation is possible about a sigma bond because the cylindrical symmetry means bond strength is unchanged by rotation.

3. PI (π) BOND

A pi bond is formed by **lateral (sideways) overlap** of p orbitals perpendicular to the internuclear axis. Electron density lies above and below (or in front and behind) the internuclear axis. Pi bonds are **NOT cylindrically symmetrical**, so rotation about a pi bond would break it.

Properties of Pi Bond:

- Formed only by lateral overlap of p-p (or d-d, p-d) orbitals
- Pi bonds are weaker than sigma bonds (less effective overlap)
- NO free rotation about a double or triple bond

- Pi electrons are more loosely held → more reactive (electrophilic attack sites)

4. MULTIPLE BONDS

Bond Type	Composition	Example	Total Bonds
Single bond	1 sigma (σ)	H-H, C-C, H-Cl	1
Double bond	1 sigma + 1 pi	O=O, C=C, C=O	2
Triple bond	1 sigma + 2 pi	N $\equiv$ N, C $\equiv$ C, C $\equiv$ O	3

5. BOND STRENGTH: $\sigma > \pi$

Sigma bonds are **stronger** than pi bonds because head-on overlap gives a larger overlap integral (more electron density between nuclei) compared to lateral overlap. Breaking a double bond requires ~611 kJ/mol (C=C), while a single C-C bond requires ~347 kJ/mol — the extra ~264 kJ is the pi bond energy.

6. COUNTING σ AND π BONDS IN MOLECULES

Molecule	Structure	σ bonds	π bonds	Total bonds
H ₂	H-H	1	0	1
HF	H-F	1	0	1
F ₂	F-F	1	0	1
O ₂	O=O	1	1	2
N ₂	N $\equiv$ N	1	2	3
CO ₂	O=C=O	2	2	4
C ₂ H ₄	Ethylene	5	1	6
C ₂ H ₂	Acetylene	3	2	5
Benzene	C ₆ H ₆	12	3	15

7. RESONANCE IN VBT

Some molecules cannot be represented by a single VBT structure. The actual structure is a **resonance hybrid** — a weighted average of all contributing structures. Resonance is indicated by a double-headed arrow ($\leftrightarrow$) between structures.

Important Resonance Examples:

- Benzene (C₆H₆): 2 Kekule structures; actual bond length 140 pm (between C-C 154pm and C=C 134pm)
- SO₃: 3 equivalent resonance structures; all S-O bonds equal (~142 pm)

- CO₃²⁻: 3 equivalent resonance structures; all C-O bonds equal (~129 pm)
- Resonance lowers the energy of the molecule (resonance stabilisation energy)
- Benzene resonance energy $\approx$ 150 kJ/mol

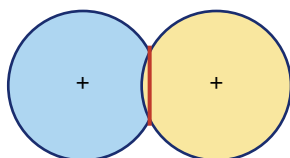
8. LIMITATIONS OF VBT

Limitations:

- Cannot explain the **paramagnetism of O₂**. VBT predicts O₂ has no unpaired electrons, but experimentally O₂ is paramagnetic (2 unpaired electrons). MO Theory explains this correctly.
- Cannot fully explain the **equal bond lengths** in benzene without invoking resonance as an artificial concept.
- Fails to predict **bond order** for species like O₂⁺, O₂⁻, O₂²⁻ directly.
- Does not satisfactorily explain **colour** and electronic spectra of transition metal complexes.
- Cannot account for the formation of coordinate/dative bonds easily.

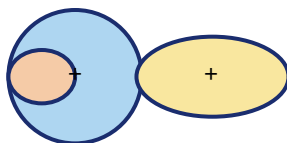
9. ORBITAL OVERLAP DIAGRAMS

Sigma Bond (s-s overlap)



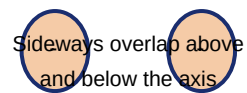
Head-on / axial overlap
Cylindrically symmetrical

Sigma Bond (s-p overlap)



s orbital overlapping with
large lobe of p orbital

Pi Bond (p-p lateral)



Sideways overlap above
and below the axis

10. WORKED EXAMPLES

Example 1 — Sigma and Pi bonds in N₂

Question: Identify the sigma and pi bonds in N₂.

Solution:

- N₂ has a triple bond: N≡N
- 1 sigma bond: formed by head-on overlap of one p orbital from each N along the internuclear axis
- 2 pi bonds: formed by lateral overlap of the remaining two p orbitals (p_x-p_x and p_y-p_y)
- Bond order = 3 (one sigma + two pi)
- Answer: N₂ has 1 σ bond and 2 π bonds. Bond energy $\approx$ 945 kJ/mol (very strong!)

Example 2 — Sigma and Pi bonds in CO₂

Question: Count the sigma and pi bonds in CO₂ (O=C=O).

Solution:

- CO₂ structure: O=C=O (two double bonds)
- Each C=O double bond = 1 sigma + 1 pi
- Total sigma bonds: 2 (one in each C=O)
- Total pi bonds: 2 (one in each C=O)
- C is sp hybridised: 2 sp orbitals form 2 sigma bonds; 2 unhybridised p orbitals form 2 pi bonds
- Answer: CO₂ has 2 σ bonds and 2 π bonds

Example 3 — Type of overlap in HF

Question: What type of orbital overlap forms the H-F bond in HF?

Solution:

- H contributes: 1s orbital (half-filled)
- F contributes: 2p orbital (half-filled, along internuclear axis)
- Overlap type: s-p head-on overlap → SIGMA bond
- The bond is formed along the internuclear axis
- Answer: HF has a sigma bond formed by s-p overlap

Example 4 — Counting bonds in benzene C₆H₆

Question: Count all sigma and pi bonds in benzene.

Solution:

- Benzene structure: 6 C-H bonds + ring with alternating C=C and C-C
- C-H bonds: 6 sigma bonds (each sp² C forms one C-H sigma bond)
- C-C sigma bonds in ring: 6 (each adjacent C pair connected by sigma)
- Pi bonds: 3 (delocalised over ring; Kekule structure shows 3 C=C)
- Total sigma bonds = 6 (C-H) + 6 (C-C) = 12
- Total pi bonds = 3
- Answer: Benzene has 12 σ bonds and 3 π bonds

Example 5 — VBT explanation for O₂ and its limitation

Question: Explain the structure of O₂ using VBT and state its limitation.

Solution:

- VBT structure of O₂: O=O (one double bond = 1 sigma + 1 pi)
- VBT predicts all electrons are paired → should be diamagnetic
- Experimental observation: O₂ is PARAMAGNETIC (attracted to magnetic field)
- Paramagnetism requires unpaired electrons
- VBT FAILS here — it cannot explain 2 unpaired electrons in O₂
- Molecular Orbital Theory correctly places 2 electrons in degenerate π^* antibonding orbitals (unpaired)
- Answer: VBT limitation — cannot explain paramagnetism of O₂

11. KEY FACTS — QUICK REVISION

KEY FACTS / QUICK REVISION — VALENCE BOND THEORY

- Sigma bond: head-on overlap; cylindrically symmetrical; free rotation possible
- Pi bond: lateral overlap; NOT cylindrically symmetrical; no free rotation
- Bond strength: $\sigma > \pi$ (sigma has larger overlap)
- Single bond = 1 sigma; Double = 1 sigma + 1 pi; Triple = 1 sigma + 2 pi
- H₂: s-s sigma | HF: s-p sigma | F₂: p-p sigma | O₂: 1sigma+1pi | N₂: 1sigma+2pi
- Benzene: 12 sigma + 3 pi bonds
- Resonance stabilises molecules; lowers energy below any single structure
- VBT limitation: cannot explain paramagnetism of O₂
- VBT limitation: bond lengths in benzene not satisfactorily explained without resonance
- Pi electrons are more exposed and more reactive than sigma electrons

12. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

1. Which of the following molecules does NOT have a pi bond?

- (A) N_2
- (B) O_2
- (C) F_2
- (D) CO_2

2. The total number of sigma bonds in ethylene (C_2H_4) is:

- (A) 3
- (B) 4
- (C) 5
- (D) 6

3. A pi bond is formed by:

- (A) Head-on overlap of s orbitals
- (B) Lateral overlap of p orbitals
- (C) Head-on overlap of p orbitals
- (D) Overlap of s and d orbitals

4. Which statement about sigma bonds is INCORRECT?

- (A) Formed by head-on overlap
- (B) Cylindrically symmetrical about bond axis
- (C) Free rotation is not possible
- (D) Stronger than pi bonds

5. In acetylene (C_2H_2), how many pi bonds are present?

(A) 0

(B) 1

(C) 2

(D) 3

6. The resonance energy of benzene is approximately:

(A) 50 kJ/mol

(B) 100 kJ/mol

(C) 150 kJ/mol

(D) 200 kJ/mol

7. Which of the following CANNOT be explained by Valence Bond Theory?

(A) Structure of H_2

(B) Paramagnetism of O_2

(C) Triple bond in N_2

(D) Double bond in O_2

8. The bond between H and F in HF is formed by:

(A) s-s overlap

(B) p-p overlap

(C) s-p overlap

(D) d-p overlap

9. How many total bonds (sigma + pi) are present in CO₂?

(A) 2

(B) 3

(C) 4

(D) 6

10. Which molecule has the highest sigma-to-pi bond ratio?

(A) N₂

(B) C₂H₄

(C) C₂H₂

(D) CO₂

13. ANSWER KEY WITH EXPLANATIONS

1. Answer: C — F₂

F₂ has a single bond (p-p head-on sigma bond only). No pi bond. N₂, O₂, CO₂ all have pi bonds.

2. Answer: C — 5 sigma bonds

C₂H₄: Each C has 2 C-H bonds (4 total C-H sigma) + 1 C-C sigma bond = 5 sigma bonds total. The C=C has 1 sigma + 1 pi.

3. Answer: B

Pi bonds are formed by lateral (sideways) overlap of p orbitals above and below the internuclear axis.

4. Answer: C — Free rotation IS possible about sigma bonds

Sigma bonds ARE cylindrically symmetrical, so free rotation IS possible. Statement C is incorrect because it says rotation is NOT possible.

5. Answer: C — 2 pi bonds

C₂H₂ has a triple bond (C≡C) = 1 sigma + 2 pi. So there are 2 pi bonds.

6. Answer: C — 150 kJ/mol

Benzene resonance energy (stabilisation energy compared to hypothetical cyclohexatriene) is approximately 150 kJ/mol.

7. Answer: B — Paramagnetism of O₂

VBT predicts O₂ has all electrons paired (diamagnetic), but experimentally O₂ is paramagnetic with 2 unpaired electrons. MO Theory explains this correctly.

8. Answer: C — s-p overlap

H contributes 1s orbital; F contributes 2p orbital along the internuclear axis. This is s-p head-on sigma overlap.

9. Answer: C — 4 bonds

CO₂ (O=C=O): Each C=O double bond = 1 sigma + 1 pi. Two C=O bonds give 2 sigma + 2 pi = 4 bonds total.

10. Answer: B — C₂H₄

C₂H₄ (ethylene): 5 sigma + 1 pi = ratio 5:1. N₂: 1:2; C₂H₂: 3:2 (ratio 1.5:1); CO₂: 2:2 (1:1). C₂H₄ has the highest sigma-to-pi ratio.